

the argument. Also assume that *removeDuplicates* can create a temporary file for handling large text files. Now, the exception safety levels for *removeDuplicates* are as follows:

No exception safety: While removing duplicates, the method can throw some exception. The temporary and the input files—if opened—might not get closed. The input file might be left corrupted, since it is partially manipulated and is not closed properly.

Basic exception safety: The temporary and the input files—if opened—would be closed. The input file might have only some of the duplicate entries removed, leaving the input text file in a partially complete state.

Strong exception safety: The temporary and the input files—if opened—would be closed. Either the input file would be left untouched, or all duplicates would have been removed. This might be implemented by initially copying the contents to a temporary file. If the duplicate removal failed in between, then the input file would remain untouched (roll-back), or else, if duplicate line removal succeeded fully, then the input

file would be replaced entirely with the contents of the temporary file (commit).

No-throw exception safety: No exceptions can get thrown out of *removeDuplicates*, and the function must always succeed—it is not possible to implement such a *removeDuplicates* method that will always succeed.

No exception safety is not acceptable in practice, as in this duplicate line removal example. We should at least provide basic exception safety; in this example, basic exception safety is not sufficient, but achieving even that is better than no exception safety. We cannot ensure no-throw exception safety for all methods—only for critical methods can it be done. Strong exception safety is the desired level for real-world applications, as illustrated in this example. **END** 

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